

terminal twigs of small shrubs such as willow and birch in winter, although why this food supply might cycle in 10-year intervals is uncertain. Second, the cycles may be due to predator-prey interactions. Many predators other than lynx eat hares, and they may overexploit their prey.

Let's consider the evidence for the two hypotheses. If hare cycles are due to winter food shortage, then the cycles should stop if extra food is provided to a field population. Researchers conducted such experiments in the Yukon for 20 years—over two hare cycles. They found that hare populations in the areas with extra food increased about threefold in density but continued to cycle in the same way as the unfed control populations. Therefore, food supplies alone do not cause the hare cycles shown in Figure 53.19, so we can reject the first hypothesis.

To study the effects of predation, ecologists used radio collars to track individual hares to determine why they died. Predators, including lynx, coyotes, hawks, and owls, killed 95% of the hares in such studies. None of the hares appeared to have died of starvation. These data support the second hypothesis. When ecologists set up electric fences to exclude predators from certain areas, the collapse in survival that normally occurs in the decline phase of the cycle was nearly eliminated. Overexploitation by predators thus seems to be an essential part of snowshoe hare cycles; without predators, it is unlikely that hare populations would cycle in northern Canada.

Immigration, Emigration, and Metapopulations

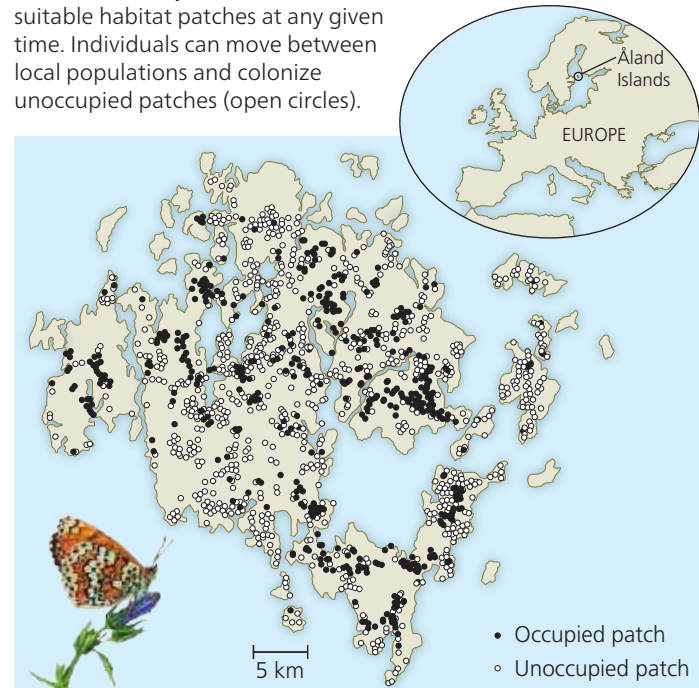
So far, our discussion of population dynamics has focused mainly on the contributions of births and deaths. However, immigration and emigration also influence populations. When a population becomes crowded and resource competition increases, emigration often increases.

Immigration and emigration are particularly important when a number of local populations are linked, forming a **metapopulation**. Local populations in a metapopulation can be thought of as occupying discrete patches of suitable habitat in a sea of otherwise unsuitable habitat. Such patches vary in size, quality, and isolation from other patches, factors that influence how many individuals move among the populations. If one population becomes extinct, the patch it occupied may be recolonized by immigrants from another population.

The Glanville fritillary (*Melitaea cinxia*) illustrates the movement of individuals between populations. On average, this butterfly is found in about 500 meadows across the Åland Islands of Finland, but its potential habitat in the islands is much larger, approximately 4,000 suitable patches. New populations of the butterfly regularly appear and existing populations become extinct, constantly shifting the locations of the 500 colonized patches (Figure 53.20). The species persists in a balance of local extinctions and recolonizations.

An individual's ability to move between populations depends on a number of factors, including its genetic

▼ **Figure 53.20 The Glanville fritillary: a metapopulation.** On the Åland Islands, local populations of this butterfly (filled circles) are found in only a fraction of the suitable habitat patches at any given time. Individuals can move between local populations and colonize unoccupied patches (open circles).



makeup. One gene that has a strong effect on the Glanville fritillary's ability to move is *Pgi*, which codes for the enzyme phosphoglucosomerase. This enzyme catalyzes the second step of glycolysis (see Figure 9.8), and its activity correlates with the rate of CO₂ production from respiration by the butterflies. Ecologists studied butterflies known to be heterozygous or homozygous for a single nucleotide polymorphism in *Pgi*. They tracked the movements of individual butterflies using radar and transponders attached to the butterflies that emit an identifying signal. Butterfly movements ranged widely, from 10 m to 4 km, in 2-hour periods. Heterozygous individuals flew more than twice as far in the morning and at lower ambient temperatures than did homozygous individuals. The results indicated a fitness advantage to the heterozygous genotype in low temperatures and a greater likelihood of heterozygotes colonizing new locations in the metapopulation.

▼ **A Glanville fritillary (*Melitaea cinxia*) wearing a tracking transponder.**

